CS 571

Homework 2 – Entity and Relation Extraction

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This homework is finished with the assistance of Codex (GPT-5.6 Sol).

I. Supervised Entity Tagging

Task and protocol. I convert mention spans to token-level BIO tags over PER, ORG, LOC, TIME, NUM, and MISC. A mention is correct only when document, sentence, start, end, and type exactly match gold. All settings are selected on the 500-document dev split; test is evaluated only after the NER and RE stages are locked.

Classifier and hand-written feature set. The tagger is trained from scratch with scikit-learn: an averaged multiclass `SGDClassifier` with one-vs-rest logistic loss over nonnegative `FeatureHasher` columns. For token w_i, my feature function emits: original and case-folded form; prefixes/suffixes of lengths 1–5; compressed word shape; capped length; uppercase, title-case, lowercase, digit, embedded-digit, hyphen, and alphanumeric flags; case-folded word and shape for a ± 3 window; boundary symbols; left/right lexical bigrams; and the preceding BIO tag (gold in training, hypothesized in decoding). No pretrained encoder, pipeline, embedding, or external NER model is used.

Training and decoding. The selected model has 2^{19} hashed dimensions, L2 $\alpha = 10^{-6}$, at most 20 epochs, tolerance 10^{-3}, averaged weights, O-token weight .60, and seed 571. A width-4 left-to-right beam uses class log probabilities and rejects an I-X transition unless preceded by B-X or I-X; a final repair converts any remaining orphan I-X to B-X. Longer gold spans win annotation overlaps. Decoded mentions are grouped iff type and case-folded, whitespace-normalized surface form match.

	Dev			Test		
Type	P	R	F1	P	R	F1
PER	.7837	.8643	.8220	.7791	.8591	.8172
ORG	.7074	.6251	.6637	.7010	.6177	.6567
LOC	.7802	.8686	.8221	.7942	.8533	.8227
TIME	.9368	.9170	.9268	.9241	.9048	.9144
NUM	.7907	.8395	.8144	.7937	.8646	.8276
MISC	.6413	.5269	.5785	.6192	.5364	.5748
Micro	**.7783**	**.7854**	**.7818**	**.7766**	**.7818**	**.7792**

Table 1. Exact mention span-and-type results for the allowed hand-feature tagger.

NER confusion matrix and errors. Table 2 aligns predictions to gold only when boundaries are exact; unmatched gold/predictions enter the Miss column/Spurious row. Thus the diagonal sums to 10,124 exact true positives and the off-diagonal cells are 1,497 type errors.

Gold \ Pred	PER	ORG	LOC	TIME	NUM	MISC	Miss
PER	**2025**	29	100	1	1	62	139
ORG	128	**1144**	240	1	3	107	229
LOC	133	81	**3520**	0	1	81	309
TIME	0	3	7	**1815**	20	7	154
NUM	1	0	0	8	**581**	0	82
MISC	133	158	185	3	4	**1039**	415
Spurious	179	217	380	136	122	382	–

Table 2. Test exact-boundary entity-type confusion counts.

ORG→LOC (240) and MISC→LOC (185) dominate type confusion; MISC also has the most misses (415). A separate overlap pairing finds 593 boundary-only errors and 351 errors involving both boundary and type. The model handles regular temporal/numeric patterns well but lexical ambiguity and unseen long names remain difficult.

II. Relation Extraction

Two-stage protocol and entity construction. NER is selected and frozen on dev before RE selection. BIO segments become mentions; equal type plus case-folded normalized surface form defines an entity cluster and its `vertexSet` index. For n entities, every ordered (h, t) with $h \neq t$ is a candidate, so direction is explicit. Self-pairs and cross-document pairs are excluded.

Feature set and classifier. For each pair, sparse categorical features encode head/tail types and their order; normalized names, name tokens, and ordered name pair; mention-count buckets; title occurrence; closest-mention sentence distance, same-sentence status and argument order; token-distance bucket; up to 12 intervening tokens; and immediate left/right argument context. They map to 2^{19} nonnegative hashed dimensions. The classifier is averaged multiclass SGD with logistic loss, L2 $\alpha = 2 \times 10^{-6}$, 25 epochs, tolerance 10^{-3}, and seed 571. All positives are retained; unrelated pairs are sampled five per positive. Class weight is $\min(8, (n_{\max}/n_y)^{0.2})$. The dev-selected multi-label decoder emits at most two relations when $p(r) \geq .90p(\texttt{no_relation})$.

Output and evaluation. The predictor preserves sentences and predicted `vertexSet`, removes input labels, and writes dictionaries `{h,t,r}`. Both final files pass `check_format.py -re` and `re_eval.py`. Pipeline evaluation aligns predicted to gold entities through exact shared spans; triples with an unaligned endpoint are false positives.

Split	Training	Entities	P	R	F1	TP	FP	FN
Dev	supervised	Pipeline	.3194	.3569	.3371	6169	13148	11115
Dev	+ distant	Pipeline	.3255	.3548	**.3395**	6133	12711	11151
Test	supervised	Pipeline	.3219	.3607	.3402	6293	13258	11155
Test	+ distant	Pipeline	.3246	.3609	**.3418**	6297	13100	11151
Test	+ distant	Gold	.4839	.4629	.4732	8077	8614	9371

Table 3. Relation micro scores with allowed NER entities.

RE confusion matrix. Because RE is multi-label, Table 4 is the micro binary confusion matrix over all 19,072,320 possible gold-entity (h, t, r) decisions (198,670 ordered pairs times 96 relations). It accompanies, rather than replaces, the pipeline score.

Gold \ Pred	Relation	No relation
Relation	**8,077** (TP)	9,371 (FN)
No relation	8,614 (FP)	**19,046,258** (TN)

Table 4. Test gold-entity micro confusion matrix.

Error analysis and pipeline-versus-gold gap. Replacing predicted entities with gold raises test F1 from .3418 to .4732, a .1314 gap. Predicted entities reach 90.1% of gold entities and make 86.9% of gold triples reachable; 5,930 predicted triples have an unaligned endpoint. Even with gold entities, 9,371 false negatives show that relation classification is still a major bottleneck. Frequent geographic labels remain confusable, while P527/P361 (has-part/part-of) suffer from direction and sparse evidence. Relation-specific thresholds and entity confidence are the clearest next steps.

III. Extra Credit: Distant Supervision

Database and weak labels. I use Wikidata because DocRED's 96 predicates are Wikidata property IDs and Wikidata represents item–property–value statements [2]. Following DocRED [1], mentions linked to the same KB item are merged and item pairs receive their Wikidata properties. For reproducibility, `prepare_distant_supervision.py` matches title plus complete tokenized text to official `train_distant.json`, rather than using changing live queries. All 10,000 documents match, restoring 192,955 weak entity clusters and 147,974 weak relation labels.

NER use and dev selection. The final model keeps the same hand-written features and scikit-learn classifier. It adds all weak entity tokens plus O tokens within two positions of a weak mention; focused sampling avoids letting roughly two million weak O tokens dominate. Weak examples receive source weight .50 versus 1.0 for human examples. The weight was chosen on dev (.10, .20, .30, .50, .75 tried); features, classifier, and decoder remain unchanged. Thus no pretrained model is introduced.

RE use and dev selection. All weak positive pairs and sampled negative pairs are included with source weight .25. Human rows retain weight 1.0. The final decoder cap 2 and ratio .90 are selected on full-pipeline dev after NER is frozen. Stronger weak weight 1.0 reduced dev F1 to .3291 and was rejected.

Stage	Split	Supervised F1	+ distant F1	Gain
NER	Dev	.7578	.7818	+.0240
NER	Test	.7514	.7792	+.0278
RE pipeline	Dev	.3371	.3395	+.0024
RE pipeline	Test	.3402	.3418	+.0016

Table 5. Distant-supervision ablation with each model's dev-selected settings.

Distant supervision clears two F1 points for NER but gives only a small RE gain; the report does not attribute decoder improvements to weak labels.

Reproducibility and output QA. `make train-ner`, `make predict-ner SPLIT=dev`, and `make eval-ner SPLIT=dev` are the one-command NER operations; RE has parallel targets. Test replaces `dev` only after selection. The four JSON files pass both format checks and evaluators. I opened the NER and RE test outputs in the supplied annotation viewer; all 500 documents rendered with populated BIO/entity and relation layers.

References

[1] Y. Yao et al. "DocRED: A Large-Scale Document-Level Relation Extraction Dataset." ACL 2019. https://aclanthology.org/P19-1074/

[2] Wikimedia Foundation. "Wikidata: Introduction" and "Wikidata: Data model." Accessed October 5, 2026. https://www.wikidata.org/wiki/Wikidata:Introduction; https://www.wikidata.org/wiki/Help:Data_model